MTH356

Timed Online Assignment – January Semester 2022

Advanced Mathematical Optimization

Tuesday, 24 May 2022 **4:00 pm – 6:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **FOUR (4)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

Consider the following nonlinear equation

$$f(x) = x^3 + \ln x - 3 = 0.$$

(a) Show that there is a root in the interval $[1, 2]$.

(3 marks)

(b) Demonstrate that the root satisfies the following two equations.

$$x = g_1(x) = \sqrt[3]{3 - \ln x}$$
$$x = g_2(x) = e^{(3-x^3)}$$

(3 marks)

(c) Two simple iteration schemes are proposed to find the root of $f(x)$ in the interval $[1.3, 1.5]$. The first scheme is based on $g_1(x)$ and the second scheme is based on $g_2(x)$. Determine if $g_1(x)$ defines a contraction mapping by checking the two conditions for a contraction mapping. Likewise, determine if $g_2(x)$ defines a contraction mapping by checking the two conditions for a contraction mapping.

(13 marks)

(d) Solve the equation, correct to three decimal places, by applying the iterative solution technique with starting point $x_0 = 1.35$.

(6 marks)

Question 2

(a) Let f be a convex differentiable function on R^n. Show that for any $x, y \in \mathbb{R}^n$

$$\left(\nabla f(x) - \nabla f(y)\right)^T (x - y) \geq 0.$$

(6 marks)

(b) Let f and g be two convex and twice continuously differentiable functions on $\mathbb{R}$. Suppose in addition that f is monotone increasing. Show that the composite function $h(x) = f(g(x))$ is convex.

(7 marks)

(c) Construct two convex functions f and g so that the composite function $h(x) = f(g(x))$ is not convex.

(7 marks)

Question 3

(a) Find all the stationary points of the function

$$f(x, y) = 3x^2y + y^3 - 3x^2 - 3y^2 + 1.$$

Determine whether they are local minima, local maxima, or saddle points.

(6 marks)

(b) Consider the function

$$f(x, y) = 25x^2 + y^2 + 5.$$

Apply one iteration of the steepest descent method with $(1, 3)^T$ as the starting point to minimize $f(x, y)$. Find $\left(x^{(1)}, y^{(1)}\right)^T$ and hence, $f(x^{(1)}, y^{(1)})$.

(9 marks)

(c) Consider the function

$$f(x, y) = x^2 + 4y^2 - 2xy + 2.$$

Apply one iteration of the Newton's method with $(1, 1)^T$ as the starting point to minimize $f(x, y)$. Find $\left(x^{(1)}, y^{(1)}\right)^T$ and hence, $f(x^{(1)}, y^{(1)})$.

(10 marks)

Question 4

Consider the following nonlinear programming problem (NLP).

$$\begin{aligned} \min\ & f(x) = -x_2^2 \\ \text{s.t. } & h_1(x) = x_1^2 - x_2 + x_3^2 \leq 0 \\ & h_2(x) = x_1^2 + x_2^2 + x_3^2 - 1 \leq 0 \end{aligned}$$

(a) Show that all feasible points are regular.

(8 marks)

(b) Write down the Karush-Kuhn-Tucker (KKT) conditions for the problem.

(5 marks)

(c) Find all the solutions satisfying the KKT conditions.

(12 marks)

(d) Determine whether the NLP has a global minimizer. Find the global minimizer if it exists.

(5 marks)

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